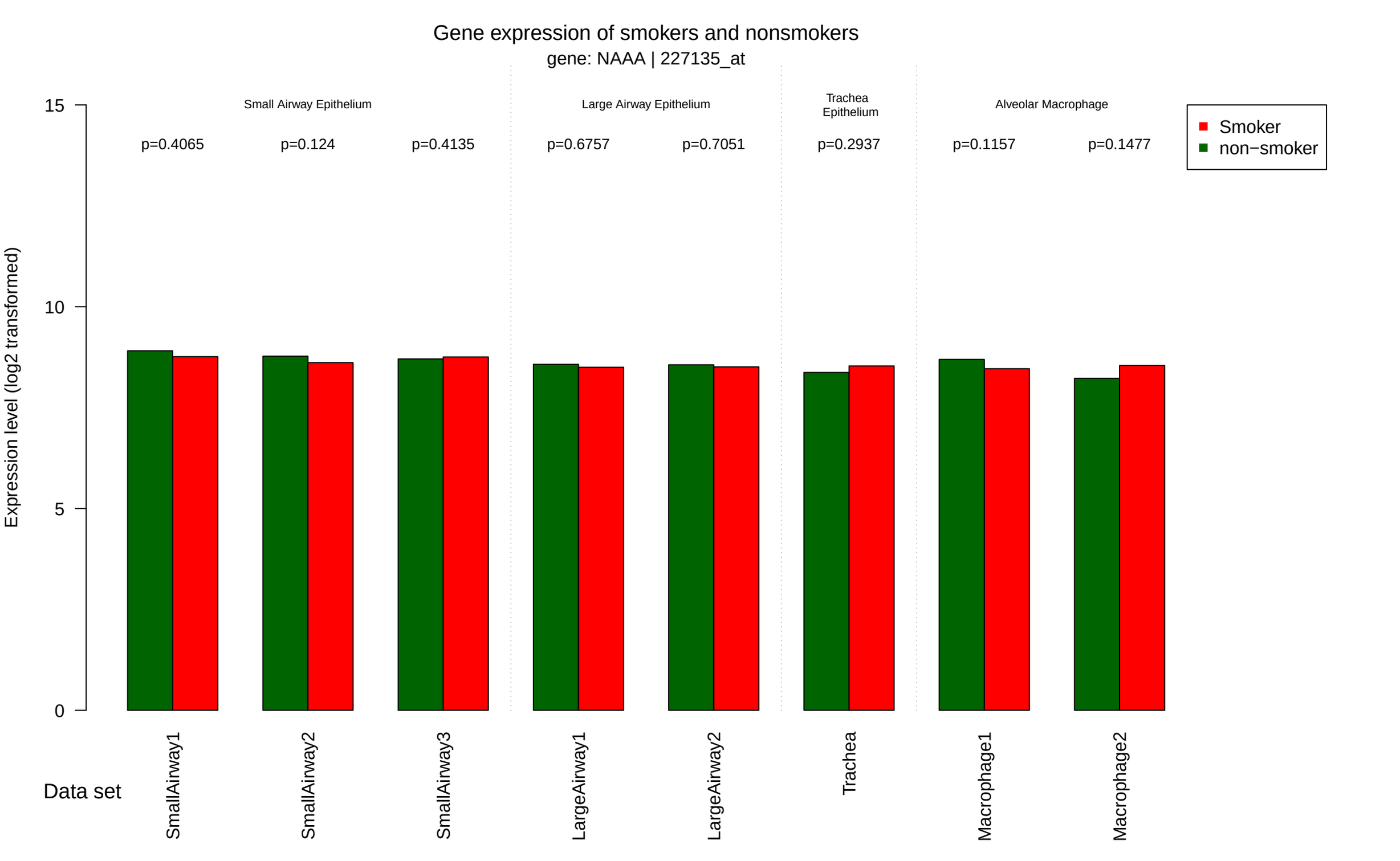
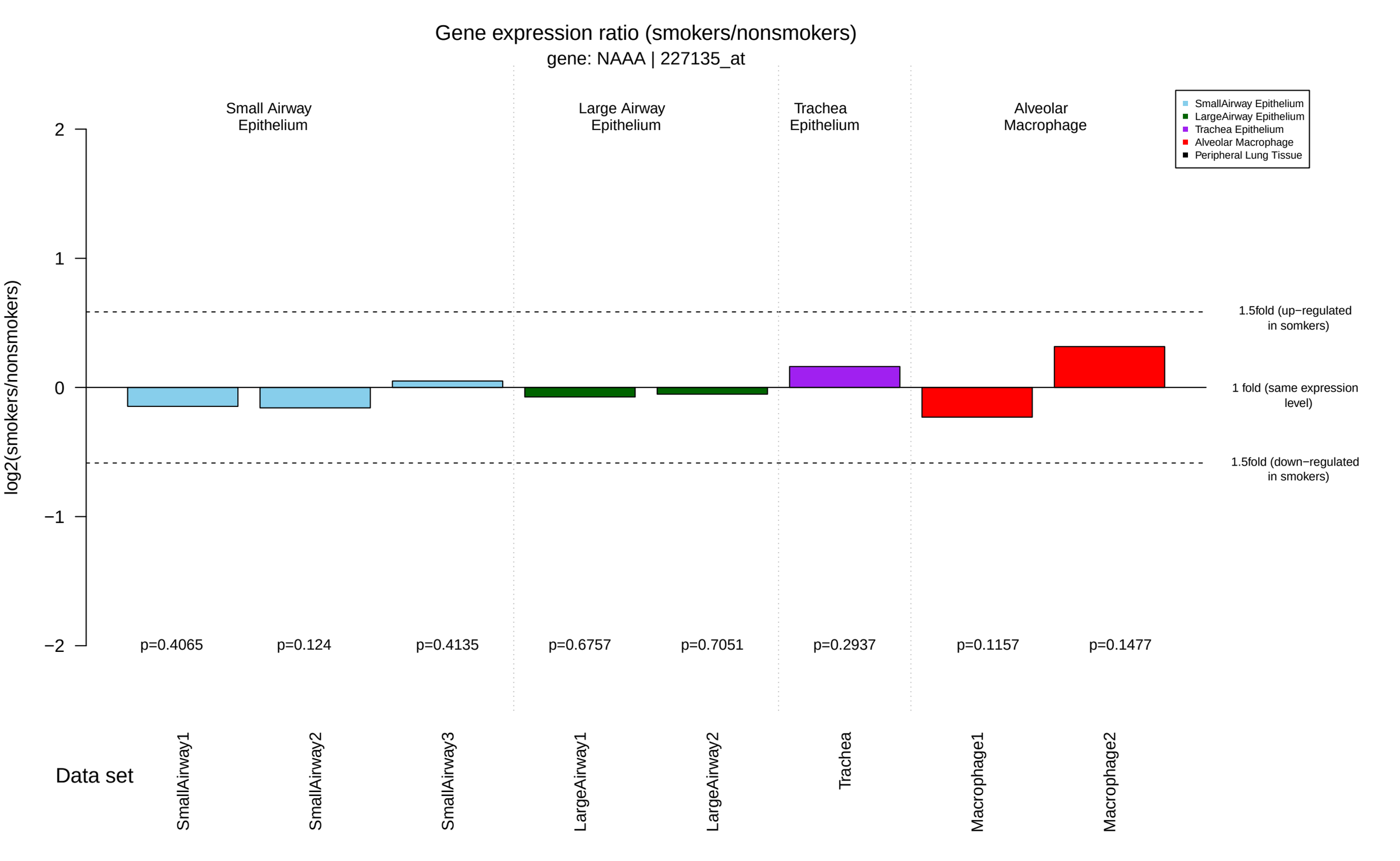
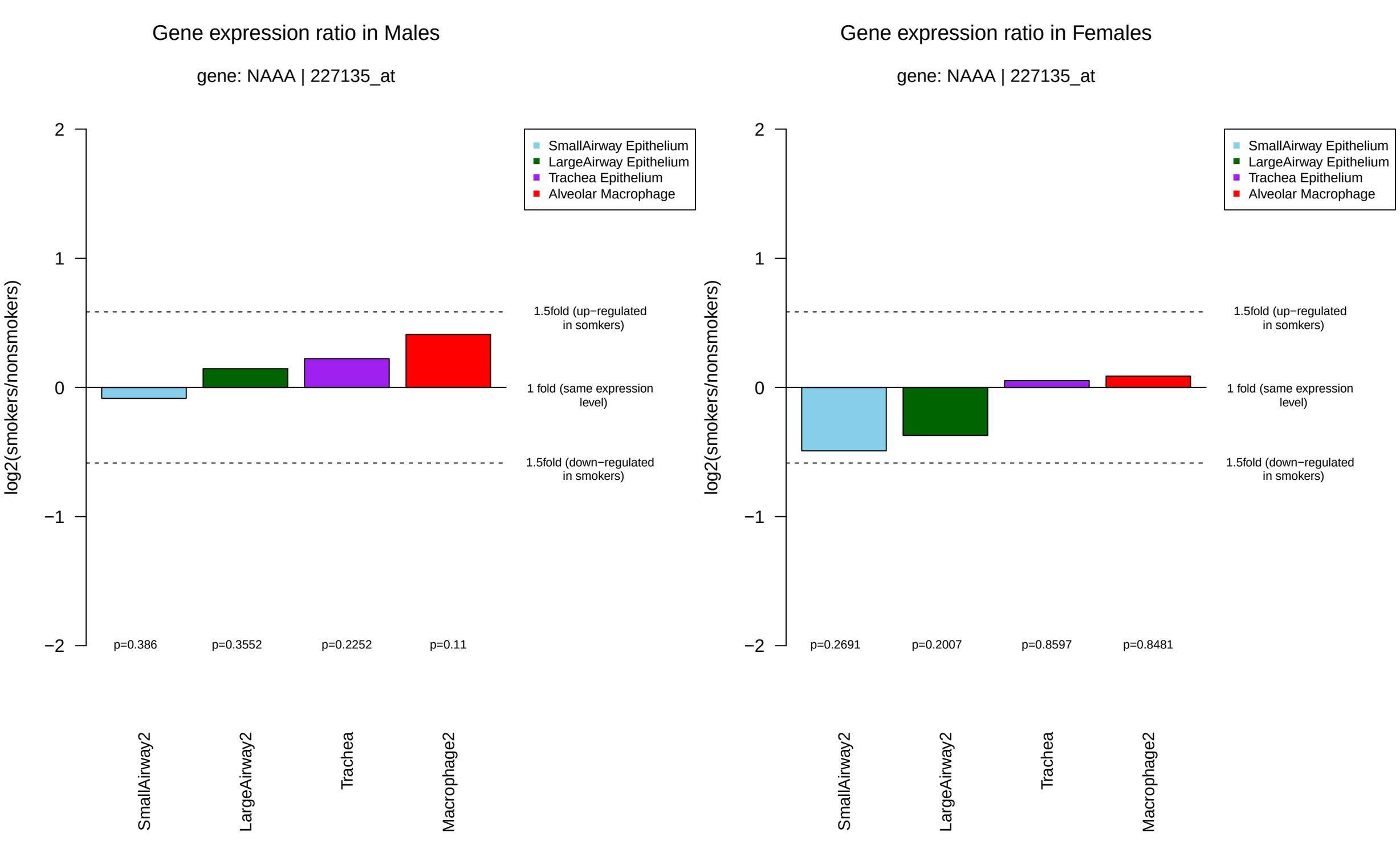
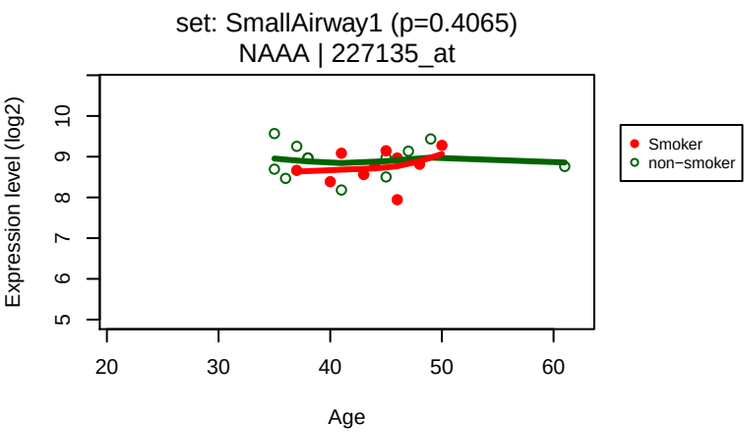
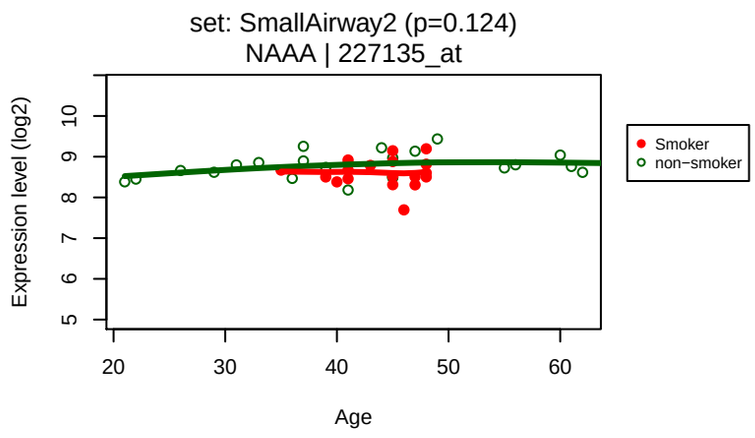
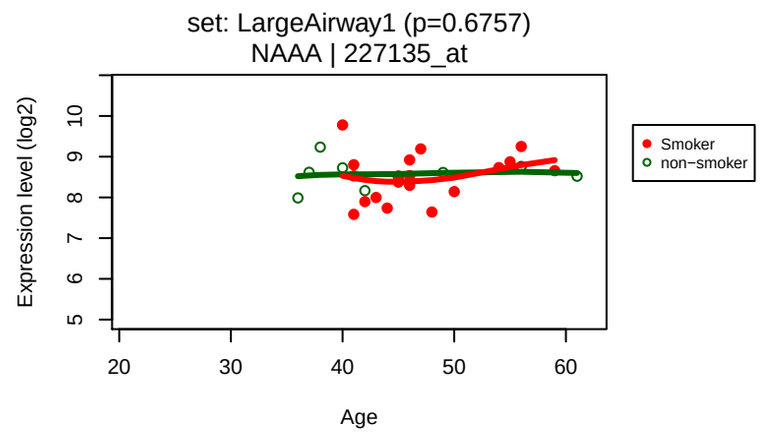

Gene expression of smokers and nonsmokers
gene: NAAA | 227135_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4065
p=0.124
p=0.4135
p=0.6757
p=0.7051
p=0.2937
p=0.1157
p=0.1477
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

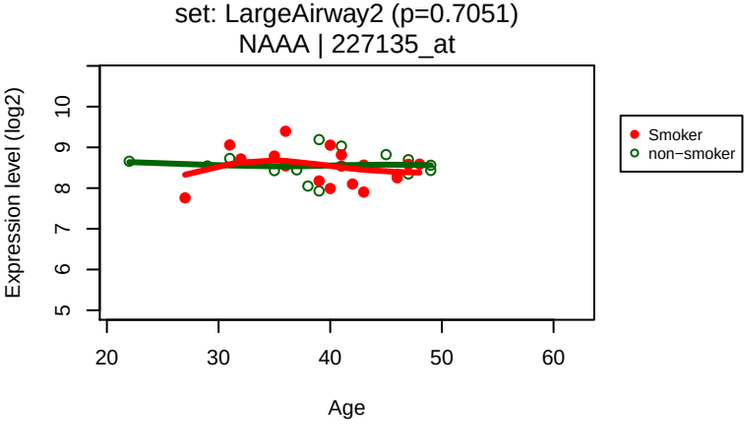

Gene expression ratio (smokers/nonsmokers)
gene: NAAA | 227135_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4065
p=0.124
p=0.4135
p=0.6757
p=0.7051
p=0.2937
p=0.1157
p=0.1477
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

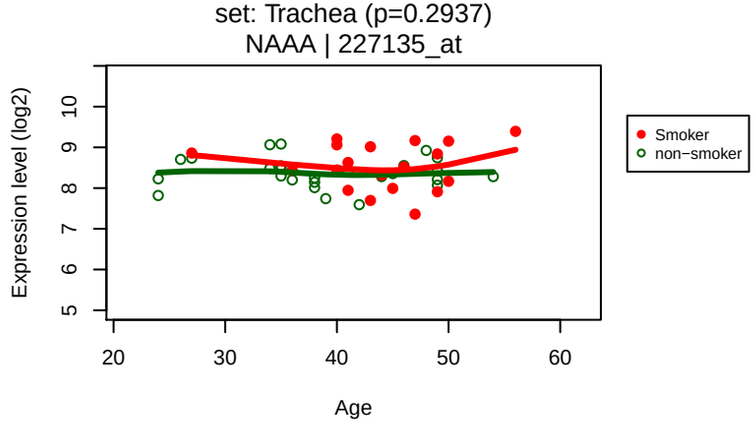

Gene expression ratio in Males
gene: NAAA | 227135_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.386
p=0.3552
p=0.2252
p=0.11
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NAAA | 227135_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2691
p=0.2007
p=0.8597
p=0.8481
SmallAirway2
LargeAirway2
Trachea
Macrophage2

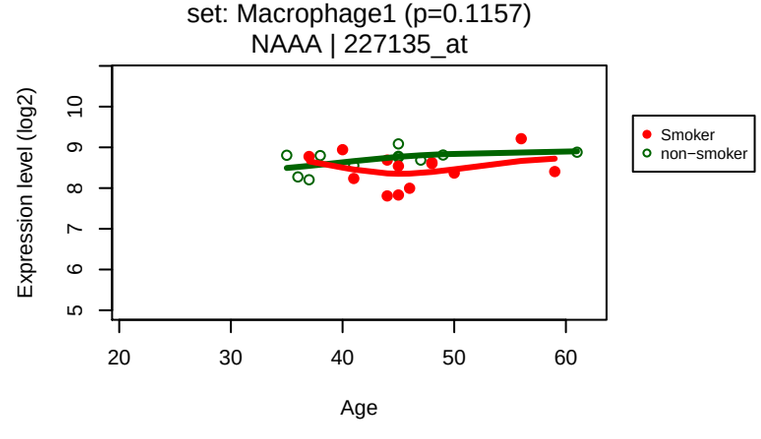

set: SmallAirway1 (p=0.4065)
NAAA | 227135_at
Expression level (log2)
Age
Smoker
non−smoker

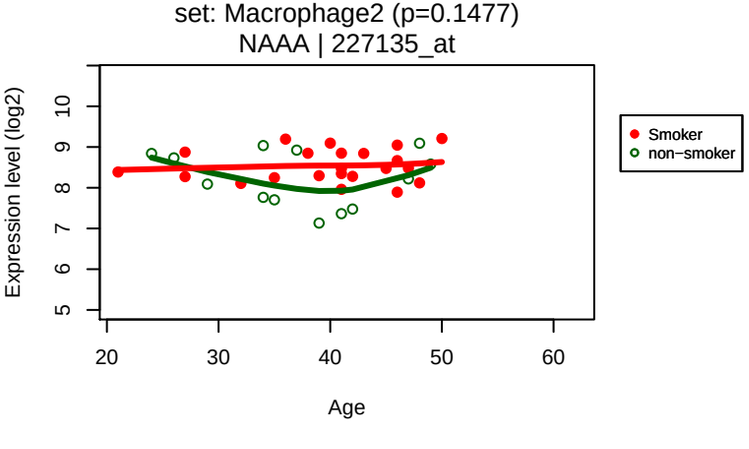

set: SmallAirway2 (p=0.124)
NAAA | 227135_at
Expression level (log2)
Age
Smoker
non−smoker

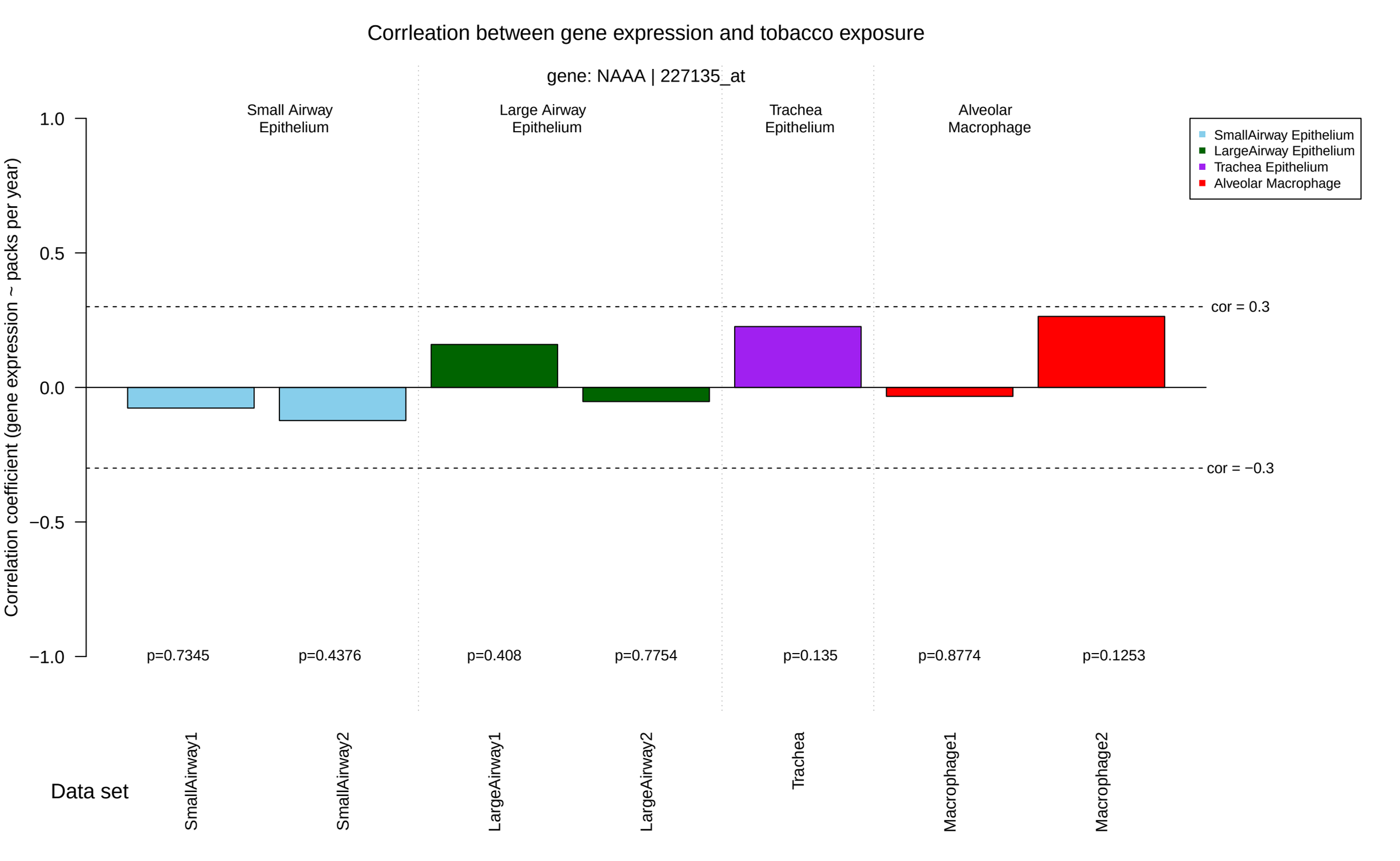

set: LargeAirway1 (p=0.6757)
NAAA | 227135_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7051)
NAAA | 227135_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2937)
NAAA | 227135_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1157)
NAAA | 227135_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1477)
NAAA | 227135_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NAAA | 227135_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7345
p=0.4376
p=0.408
p=0.7754
p=0.135
p=0.8774
p=0.1253
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2